Research Activity: Evaluating Computational Models, Machines, Formalisms, Ontologies, and Algorithms

Topics; Theory of Computation / Advanced Data Structures and Algorithms / Computer Science Foundations

Research Type: Comparative Analytical Research

Duration: 4–5 Weeks

Output: Research Paper, Presentation, and Comparative Evaluation Framework

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Background

Computer science offers numerous ways of representing computation, information processing, and problem solving. Different computational models, machines, formalisms, abstract data types (ADTs), and algorithms exhibit varying strengths, limitations, complexity characteristics, expressiveness, and practical applicability.

The purpose of this research is to investigate and evaluate a selected set of computational approaches using a rigorous analytical framework.

Learning Outcomes

Upon completion of this activity, students should be able to:

1. Differentiate computational models, machines, and formalisms, or ADTs, and algorithms.
2. Analyze theoretical and practical properties of computational approaches.
3. Apply evaluation criteria systematically.
4. Compare alternative approaches to solving the same problem.
5. Develop evidence-based recommendations.

6. Communicate research findings professionally.

Research Tasks

Part 1: Selection of Research Topic

You may refer also to this list: (https://docs.google.com/document/d/1ytVCwanIHNkiVuN7QkdKolzHJNvIx4vGfVo0bk0UXZs/edit?tab=t.0)

Choose one category:

Category A – Computational Models

Examples:

- Finite State Machine
- Pushdown Automaton
- Turing Machine
- Lambda Calculus
- Cellular Automata
- Petri Nets

Category B – Formalisms

Examples:

- Rhetorical Structure Theory
- Attribute Grammars
- Head-Driven Phrase Structure Grammar
- Temporal Logic
- Transduction Grammars

Category C – Ontologies

Examples:

- VerbNet
- Wordnet
- Stanford OWL (Web Ontology Language) (Tool: Protege)

Category D – Algorithms

Examples:

- Local Search and Physics-inspired Algorithms
    - Simulated Annealing
    - Tabu Search

- **Swarm Intelligence**
    - Ant Colony Optimization
    - Particle Swarm Optimization

Part 2: Research Question Formulation

Students must formulate:

Main Question

Example:

Which computational model provides the best balance of expressiveness, computational power, and implementation complexity for language recognition tasks?

Supporting Questions

1. What problem domain is addressed?
2. What assumptions does the approach make?
3. What are its strengths and weaknesses?
4. What are its complexity characteristics?
5. How scalable is the approach?
6. What practical applications exist?

Part 3: Literature Review

Review at least:

- 10 peer-reviewed papers
- 2 textbooks
- 2 conference publications

Part 4: Deliverables

Activity	Milestone	Deadline
Topic Selection	Submission of Topics	Jul 3, 2026
	Presentation of Abstract or Overview of Topic	Jul 6, 2026
Literature Review	Submission of Literature Review Document	Jul 24, 2026
	Presentation of Literature Review	Jul 27, 2026
Final Paper and Presentation	Submission of Final Paper	Aug 7, 2026
	Presentation of Final Paper	Aug 10, 2026